7

Model Reduction by Balanced Truncation

In this chapter we consider the problem of reducing the order of a linear multivariable dynamical system. There are many ways to reduce the order of a dynamical system. However, we shall study only two of them: balanced truncation method and Hankel norm approximation method. This chapter focuses on the balanced truncation method while the next chapter studies the Hankel norm approximation method.

A model order reduction problem can in general be stated as follows: Given a full order model $G(s)$, find a lower order model, say, an r-th order model G_r, such that G and G_r are close in some sense. Of course, there are many ways to define the closeness of an approximation. For example, one may desire that the reduced model be such that

$$G = G_r + \Delta_a$$

and Δ_a is small in some norm. This model reduction is usually called an *additive* model reduction problem. On the other hand, one may also desire that the approximation be in relative form

$$G_r = G(I + \Delta_{rel})$$

so that Δ_{rel} is small in some norm. This is called a *relative* model reduction problem. We shall be only interested in $\mathcal{L}_\infty$ norm approximation in this book. Once the norm is chosen, the additive model reduction problem can be formulated as

$$\inf_{\deg(G_r)\leq r} \|G - G_r\|_\infty$$

and the relative model reduction problem can be formulated as

$$\inf_{\deg(G_r)\le r} \left\| G^{-1}(G-G_r) \right\|_\infty$$

if G is invertible. In general, a practical model reduction problem is inherently frequency weighted, i.e., the requirement on the approximation accuracy at one frequency range can be drastically different from the requirement at another frequency range. These problems can in general be formulated as frequency weighted model reduction problems

$$\inf_{\deg(G_r)\le r} \left\| W_o(G-G_r)W_i \right\|_\infty$$

with appropriate choice of W_i and W_o. We shall see in this chapter how the balanced realization can give an effective approach to the above model reduction problems.

7.1 Model Reduction by Balanced Truncation

Consider a stable system $G \in \mathcal{RH}_\infty$ and suppose $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ is a balanced realization, i.e., its controllability and observability Gramians are equal and diagonal. Denote the balanced Gramians by Σ, then

$$A\Sigma + \Sigma A^* + BB^* = 0 \tag{7.1}$$

$$A^*\Sigma + \Sigma A + C^*C = 0. \tag{7.2}$$

Now partition the balanced Gramian as $\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix}$ and partition the system accordingly as

$$G = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right].$$

Then (7.1) and (7.2) can be written in terms of their partitioned matrices as

$$\begin{aligned}
A_{11}\Sigma_1 + \Sigma_1 A_{11}^* + B_1B_1^* &= 0 && (7.3)\\
\Sigma_1 A_{11} + A_{11}^*\Sigma_1 + C_1^*C_1 &= 0 && (7.4)\\
A_{21}\Sigma_1 + \Sigma_2 A_{12}^* + B_2B_1^* &= 0 && (7.5)\\
\Sigma_2 A_{21} + A_{12}^*\Sigma_1 + C_2^*C_1 &= 0 && (7.6)\\
A_{22}\Sigma_2 + \Sigma_2 A_{22}^* + B_2B_2^* &= 0 && (7.7)\\
\Sigma_2 A_{22} + A_{22}^*\Sigma_2 + C_2^*C_2 &= 0. && (7.8)
\end{aligned}$$

The following theorem characterizes the properties of these subsystems.

Theorem 7.1 *Assume that Σ_1 and Σ_2 have no diagonal entries in common. Then both subsystems (A_{ii}, B_i, C_i), $i = 1, 2$ are asymptotically stable.*

Proof. It is clearly sufficient to show that A_{11} is asymptotically stable. The proof for the stability of A_{22} is similar.

By Lemma 3.20 or Lemma 3.21, Σ_1 can be assumed to be positive definite without loss of generality. Then it is obvious that $\lambda_i(A_{11}) \leq 0$ by Lemma 3.19. Assume that A_{11} is not asymptotically, then there exists an eigenvalue at $j\omega$ for some ω. Let V be a basis matrix for $\mathrm{Ker}(A_{11} - j\omega I)$. Then we have

$$(A_{11} - j\omega I)V = 0 \tag{7.9}$$

which gives

$$V^*(A_{11}^* + j\omega I) = 0.$$

Equations (7.3) and (7.4) can be rewritten as

$$(A_{11} - j\omega I)\Sigma_1 + \Sigma_1(A_{11}^* + j\omega I) + B_1B_1^* = 0 \tag{7.10}$$

$$\Sigma_1(A_{11} - j\omega I) + (A_{11}^* + j\omega I)\Sigma_1 + C_1^*C_1 = 0. \tag{7.11}$$

Multiplication of (7.11) from the right by V and from the left by V^* gives $V^*C_1^*C_1V = 0$, which is equivalent to

$$C_1V = 0.$$

Multiplication of (7.11) from the right by V now gives

$$(A_{11}^* + j\omega I)\Sigma_1 V = 0.$$

Analogously, first multiply (7.10) from the right by $\Sigma_1 V$ and from the left by $V^*\Sigma_1$ to obtain

$$B_1^*\Sigma_1 V = 0.$$

Then multiply (7.10) from the right by $\Sigma_1 V$ to get

$$(A_{11} - j\omega I)\Sigma_1^2 V = 0.$$

It follows that the columns of $\Sigma_1^2 V$ are in $\mathrm{Ker}(A_{11} - j\omega I)$. Therefore, there exists a matrix $\bar{\Sigma}_1$ such that

$$\Sigma_1^2 V = V\bar{\Sigma}_1^2.$$

Since $\bar{\Sigma}_1^2$ is the restriction of Σ_1^2 to the space spanned by V, it follows that it is possible to choose V such that $\bar{\Sigma}_1^2$ is diagonal. It is then also possible to choose $\bar{\Sigma}_1$ diagonal and such that the diagonal entries of $\bar{\Sigma}_1$ are a subset of the diagonal entries of Σ_1.

Multiply (7.5) from the right by $\Sigma_1 V$ and (7.6) by V to get

$$\begin{aligned} A_{21}\Sigma_1^2 V + \Sigma_2 A_{12}^*\Sigma_1 V &= 0 \\ \Sigma_2 A_{21} V + A_{12}^*\Sigma_1 V &= 0 \end{aligned}$$

which gives

$$(A_{21}V)\bar{\Sigma}_1^2 = \Sigma_2^2(A_{21}V).$$

This is a Sylvester equation in $(A_{21}V)$. Because $\bar{\Sigma}_1^2$ and Σ_2^2 have no diagonal entries in common it follows from Lemma 2.7 that

$$A_{21}V = 0 \tag{7.12}$$

is the unique solution. Now (7.12) and (7.9) implies that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} V \\ 0 \end{bmatrix} = j\omega \begin{bmatrix} V \\ 0 \end{bmatrix}$$

which means that the A-matrix of the original system has an eigenvalue at $j\omega$. This contradicts the fact that the original system is asymptotically stable. Therefore A_{11} must be asymptotically stable. $\square$

Corollary 7.2 *If Σ has distinct singular values, then every subsystem is asymptotically stable.*

The stability condition in Theorem 7.1 is only sufficient. For example,

$$\frac{(s-1)(s-2)}{(s+1)(s+2)} = \left[\begin{array}{cc|c} -2 & -2.8284 & -2 \\ 0 & -1 & -1.4142 \\ \hline 2 & 1.4142 & 1 \end{array}\right]$$

is a balanced realization with $\Sigma = I$ and every subsystem of the realization is stable. On the other hand,

$$\frac{s^2 - s + 2}{s^2 + s + 2} = \left[\begin{array}{cc|c} -1 & 1.4142 & 1.4142 \\ -1.4142 & 0 & 0 \\ \hline -1.4142 & 0 & 1 \end{array}\right]$$

is also a balanced realization with $\Sigma = I$ but one of the subsystems is not stable.

Theorem 7.3 *Suppose $G(s) \in \mathcal{RH}_\infty$ and*

$$G(s) = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right]$$

is a balanced realization with Gramian $\Sigma = \mathrm{diag}(\Sigma_1, \Sigma_2)$

$$\begin{aligned} \Sigma_1 &= diag(\sigma_1 I_{s_1}, \sigma_2 I_{s_2}, \ldots, \sigma_r I_{s_r}) \\ \Sigma_2 &= diag(\sigma_{r+1} I_{s_{r+1}}, \sigma_{r+2} I_{s_{r+2}}, \ldots, \sigma_N I_{s_N}) \end{aligned}$$

and

$$\sigma_1 > \sigma_2 > \cdots > \sigma_r > \sigma_{r+1} > \sigma_{r+2} > \cdots > \sigma_N$$

where σ_i has multiplicity s_i, $i = 1, 2, \ldots, N$ and $s_1 + s_2 + \cdots + s_N = n$. Then the truncated system

$$G_r(s) = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is balanced and asymptotically stable. Furthermore

$$\|G(s) - G_r(s)\|_\infty \le 2(\sigma_{r+1} + \sigma_{r+2} + \cdots + \sigma_N)$$

and the bound is achieved if $r = N - 1$, i.e.,

$$\|G(s) - G_{N-1}(s)\|_\infty = 2\sigma_N.$$

Proof. The stability of G_r follows from Theorem 7.1. We shall now prove the error bound for the case $s_i = 1$ for all i. The case where the multiplicity of σ_i is not equal to one is more complicated and an alternative proof is given in the next chapter. Hence, we assume $s_i = 1$ and $N = n$.

Let

$$\begin{aligned} \phi(s) &:= (sI - A_{11})^{-1} \\ \psi(s) &:= sI - A_{22} - A_{21}\phi(s)A_{12} \\ \tilde{B}(s) &:= A_{21}\phi(s)B_1 + B_2 \\ \tilde{C}(s) &:= C_1\phi(s)A_{12} + C_2 \end{aligned}$$

then using the partitioned matrix results of section 2.3,

$$\begin{aligned} G(s) - G_r(s) &= C(sI - A)^{-1}B - C_1\phi(s)B_1 \\ &= \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} sI - A_{11} & -A_{12} \\ -A_{21} & sI - A_{22} \end{bmatrix}^{-1} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} - C_1\phi(s)B_1 \\ &= \tilde{C}(s)\psi^{-1}(s)\tilde{B}(s) \end{aligned}$$

computing this quantity on the imaginary axis to get

$$\overline{\sigma}\left[G(j\omega) - G_r(j\omega)\right] = \lambda_{\max}^{1/2}\left[\psi^{-1}(j\omega)\tilde{B}(j\omega)\tilde{B}^*(j\omega)\psi^{-*}(j\omega)\tilde{C}^*(j\omega)\tilde{C}(j\omega)\right]. \quad (7.13)$$

Expressions for $\tilde{B}(j\omega)\tilde{B}^*(j\omega)$ and $\tilde{C}^*(j\omega)\tilde{C}(j\omega)$ are obtained by using the partitioned form of the internally balanced Gramian equations (7.3)–(7.8).

An expression for $\tilde{B}(j\omega)\tilde{B}^*(j\omega)$ is obtained by using the definition of $B(s)$, substituting for $B_1B_1^*$, $B_1B_2^*$ and $B_2B_2^*$ from the partitioned form of the Gramian equations (7.3)–(7.5), we get

$$\tilde{B}(j\omega)\tilde{B}^*(j\omega) = \psi(j\omega)\Sigma_2 + \Sigma_2\psi^*(j\omega).$$

The expression for $\tilde{C}^*(j\omega)\tilde{C}(j\omega)$ is obtained analogously and is given by

$$\tilde{C}^*(j\omega)\tilde{C}(j\omega) = \Sigma_2\psi(j\omega) + \psi^*(j\omega)\Sigma_2.$$

These expressions for $\tilde{B}(j\omega)\tilde{B}^*(j\omega)$ and $\tilde{C}^*(j\omega)\tilde{C}(j\omega)$ are then substituted into (7.13) to obtain

$$\overline{\sigma}\left[G(j\omega) - G_r(j\omega)\right] = \lambda_{\max}^{1/2}\left\{\left[\Sigma_2 + \psi^{-1}(j\omega)\Sigma_2\psi^*(j\omega)\right]\left[\Sigma_2 + \psi^{-*}(j\omega)\Sigma_2\psi(j\omega)\right]\right\}.$$

Now consider one-step order reduction, i.e., $r = n-1$, then $\Sigma_2 = \sigma_n$ and

$$\overline{\sigma}\left[G(j\omega) - G_r(j\omega)\right] = \sigma_n\lambda_{\max}^{1/2}\left\{\left[1 + \Theta^{-1}(j\omega)\right]\left[1 + \Theta(j\omega)\right]\right\} \tag{7.14}$$

where $\Theta := \psi^{-*}(j\omega)\psi(j\omega) = \Theta^{-*}$ is an "all pass" scalar function. (This is the only place we need the assumption of $s_i = 1$) Hence $|\Theta(j\omega)| = 1$.

Using triangle inequality we get

$$\overline{\sigma}\left[G(j\omega) - G_r(j\omega)\right] \le \sigma_n\left[1 + |\Theta(j\omega)|\right] = 2\sigma_n. \tag{7.15}$$

This completes the bound for $r = n-1$.

The remainder of the proof is achieved by using the order reduction by one step results and by noting that $G_k(s) = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$ obtained by the "k-th" order partitioning is internally balanced with balanced Gramian given by

$$\Sigma_1 = \mathrm{diag}(\sigma_1 I_{s_1}, \sigma_2 I_{s_2}, \ldots, \sigma_k I_{s_k}).$$

Let $E_k(s) = G_{k+1}(s) - G_k(s)$ for $k = 1, 2, \ldots, N-1$ and let $G_N(s) = G(s)$. Then

$$\overline{\sigma}\left[E_k(j\omega)\right] \le 2\sigma_{k+1}$$

since $G_k(s)$ is a reduced order model obtained from the internally balanced realization of $G_{k+1}(s)$ and the bound for one-step order reduction, (7.15) holds.

Noting that

$$G(s) - G_r(s) = \sum_{k=r}^{N-1} E_k(s)$$

by the definition of $E_k(s)$, we have

$$\overline{\sigma}\left[G(j\omega) - G_r(j\omega)\right] \le \sum_{k=r}^{N-1}\overline{\sigma}\left[E_k(j\omega)\right] \le 2\sum_{k=r}^{N-1}\sigma_{k+1}.$$

This is the desired upper bound.

To see that the bound is actually achieved when $r = N-1$, we note that $\Theta(0) = I$. Then the right hand side of (7.14) is $2\sigma_N$ at $\omega = 0$. □

The singular values σ_i are called the Hankel singular values. A useful consequence of the above theorem is the following corollary.

Corollary 7.4 *Let $\sigma_i, i = 1, \ldots, N$ be the Hankel singular values of $G(s) \in \mathcal{RH}_\infty$. Then*

$$\|G(s) - G(\infty)\|_\infty \leq 2(\sigma_1 + \ldots + \sigma_N).$$

The above bound can be tight for some systems. For example, consider an n-th order transfer function

$$G(s) = \sum_{j=1}^{n} \frac{\alpha^{2j}}{s + \alpha^{2j}},$$

with $\alpha > 0$. Then $\|G(s)\|_\infty = G(0) = n$ and $G(s)$ has the following state space realization

$$G = \left[\begin{array}{cccc|c} -\alpha^2 & & & & \alpha \\ & -\alpha^4 & & & \alpha^2 \\ & & \ddots & & \vdots \\ & & & -\alpha^{2n} & \alpha^n \\ \hline \alpha & \alpha^2 & \cdots & \alpha^n & 0 \end{array}\right]$$

and the controllability and observability Gramians of the realization are given by

$$P = Q = \left[\frac{\alpha^{i+j}}{\alpha^{2i} + \alpha^{2j}}\right]$$

and $P = Q \to \frac{1}{2} I_n$ as $\alpha \to \infty$. So the Hankel singular values $\sigma_j \to \frac{1}{2}$ and $2(\sigma_1 + \sigma_2 + \cdots + \sigma_n) \to n = \|G(s)\|_\infty$ as $\alpha \to \infty$.

The model reduction bound can also be loose for systems with Hankel singular values close to each other. For example, consider the balanced realization of a fourth order system

$$G(s) = \left[\begin{array}{cccc|c} -19.9579 & -5.4682 & 9.6954 & 0.9160 & -6.3180 \\ 5.4682 & 0 & 0 & 0.2378 & 0.0020 \\ -9.6954 & 0 & 0 & -4.0051 & -0.0067 \\ 0.9160 & -0.2378 & 4.0051 & -0.0420 & 0.2893 \\ \hline -6.3180 & -0.0020 & 0.0067 & 0.2893 & 0 \end{array}\right]$$

with Hankel singular values given by

$$\sigma_1 = 1, \quad \sigma_2 = 0.9977, \quad \sigma_3 = 0.9957, \sigma_4 = 0.9952.$$

The approximation errors and the estimated bounds are listed in the following table. The table shows that the actual error for an r-th order approximation is almost the same as $2\sigma_{r+1}$ which would be the estimated bound if we regard $\sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_4$. In general, it is not hard to construct an n-th order system so that the r-th order balanced model reduction error is approximately $2\sigma_{r+1}$ but the error bound is arbitrarily close to $2(n-r)\sigma_{r+1}$. One method to construct such system is as follows: Let $G(s)$ be a stable all-pass function, i.e., $G(s)^\sim G(s) = I$, then there is a balanced realization for G so that the controllability and observability Gramians are $P = Q = I$. Next make a very small perturbation to the balanced realization then the perturbed system has a balanced realization with distinct singular values and $P = Q \approx I$. This perturbed system will have the desired properties and this is exactly how the above example is constructed.

r	0	1	2	3
$\|G - G_r\|_\infty$	2	1.996	1.991	1.9904
Bounds: $2\sum_{i=r+1}^{4} \sigma_i$	7.9772	5.9772	3.9818	1.9904
$2\sigma_{r+1}$	2	1.9954	1.9914	1.9904

7.2 Frequency-Weighted Balanced Model Reduction

This section considers the extension of the balanced truncation method to frequency weighted case. Given the original full order model $G \in \mathcal{RH}_\infty$, the input weighting matrix $W_i \in \mathcal{RH}_\infty$ and the output weighting matrix $W_o \in \mathcal{RH}_\infty$, our objective is to find a lower order model G_r such that

$$\|W_o(G - G_r)W_i\|_\infty$$

is made as small as possible. Assume that G, W_i, and W_o have the following state space realizations

$$G = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right], \quad W_i = \left[\begin{array}{c|c} A_i & B_i \\ \hline C_i & D_i \end{array}\right], \quad W_o = \left[\begin{array}{c|c} A_o & B_o \\ \hline C_o & D_o \end{array}\right]$$

with $A \in \mathbb{R}^{n\times n}$. Note that there is no loss of generality in assuming $D = G(\infty) = 0$ since otherwise it can be eliminated by replacing G_r with $D + G_r$.

Now the state space realization for the weighted transfer matrix is given by

$$W_oGW_i = \left[\begin{array}{ccc|c} A & 0 & BC_i & BD_i \\ B_oC & A_o & 0 & 0 \\ 0 & 0 & A_i & B_i \\ \hline D_oC & C_o & 0 & 0 \end{array}\right] =: \left[\begin{array}{c|c} \bar{A} & \bar{B} \\ \hline \bar{C} & 0 \end{array}\right].$$

Let $\bar{P}$ and $\bar{Q}$ be the solutions to the following Lyapunov equations

$$\bar{A}\bar{P} + \bar{P}\bar{A}^* + \bar{B}\bar{B}^* = 0 \tag{7.16}$$

$$\bar{Q}\bar{A} + \bar{A}^*\bar{Q} + \bar{C}^*\bar{C} = 0. \tag{7.17}$$

Then the input weighted Gramian P and the output weighted Gramian Q are defined by

$$P := \begin{bmatrix} I_n & 0 \end{bmatrix} \bar{P} \begin{bmatrix} I_n \\ 0 \end{bmatrix}, \quad Q := \begin{bmatrix} I_n & 0 \end{bmatrix} \bar{Q} \begin{bmatrix} I_n \\ 0 \end{bmatrix}.$$

It can be shown easily that P and Q satisfy the following lower order equations

$$\begin{bmatrix} A & BC_i \\ 0 & A_i \end{bmatrix} \begin{bmatrix} P & P_{12} \\ P_{12}^* & P_{22} \end{bmatrix} + \begin{bmatrix} P & P_{12} \\ P_{12}^* & P_{22} \end{bmatrix} \begin{bmatrix} A & BC_i \\ 0 & A_i \end{bmatrix}^* + \begin{bmatrix} BD_i \\ B_i \end{bmatrix} \begin{bmatrix} BD_i \\ B_i \end{bmatrix}^* = 0 \tag{7.18}$$

$$\begin{bmatrix} Q & Q_{12} \\ Q_{12}^* & Q_{22} \end{bmatrix} \begin{bmatrix} A & 0 \\ B_oC & A_o \end{bmatrix} + \begin{bmatrix} A & 0 \\ B_oC & A_o \end{bmatrix}^* \begin{bmatrix} Q & Q_{12} \\ Q_{12}^* & Q_{22} \end{bmatrix} + \begin{bmatrix} C^*D_o^* \\ C_o^* \end{bmatrix} \begin{bmatrix} C^*D_o^* \\ C_o^* \end{bmatrix}^* = 0. \tag{7.19}$$

The computation can be further reduced if $W_i = I$ or $W_o = I$. In the case of $W_i = I$, P can be obtained from

$$PA^* + AP + BB^* = 0 \tag{7.20}$$

while in the case of $W_o = I$, Q can be obtained from

$$QA + A^*Q + C^*C = 0. \tag{7.21}$$

Now let T be a nonsingular matrix such that

$$TPT^* = (T^{-1})^*QT^{-1} = \begin{bmatrix} \Sigma_1 & \\ & \Sigma_2 \end{bmatrix}$$

(i.e., balanced) with $\Sigma_1 = \operatorname{diag}(\sigma_1 I_{s_1}, \ldots, \sigma_r I_{s_r})$ and $\Sigma_2 = \operatorname{diag}(\sigma_{r+1} I_{s_{r+1}}, \ldots, \sigma_n I_{s_n})$ and partition the system accordingly as

$$\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & 0 \end{array}\right] = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & 0 \end{array}\right].$$

Then a reduced order model G_r is obtained as

$$G_r = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & 0 \end{array}\right].$$

Unfortunately, there is generally no known a priori error bound for the approximation error and the reduced order model G_r is not guaranteed to be stable either.

7.3 Relative and Multiplicative Model Reductions

A very special frequency weighted model reduction problem is the relative error model reduction problem where the objective is to find a reduced order model G_r so that

$$G_r = G(I + \Delta_{rel})$$

and $\|\Delta_{rel}\|_\infty$ is made as small as possible. Δ_{rel} is usually called the *relative error*. In the case where G is square and invertible, this problem can be simply formulated as

$$\min_{\deg G_r \le r} \left\| G^{-1}(G - G_r) \right\|_\infty .$$

Of course the dual approximation problem

$$G_r = (I + \Delta_{rel})G$$

can be obtained by taking the transpose of G. We will show below that, as a bonus, the approximation G_r obtained below also serves as a multiplicative approximation:

$$G = G_r(I + \Delta_{mul})$$

where Δ_{mul} is usually called the *multiplicative error*.

Theorem 7.5 *Let $G, G^{-1} \in \mathcal{RH}_\infty$ be an n-th order square transfer matrix with a state space realization*

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right].$$

Let $W_i = I$ *and* $W_o = G^{-1}(s) = \left[\begin{array}{c|c} A - BD^{-1}C & -BD^{-1} \\ \hline D^{-1}C & D^{-1} \end{array}\right].$

(a) Then the weighted Gramians P and Q for the frequency weighted balanced realization of G can be obtained as

$$PA^* + AP + BB^* = 0 \tag{7.22}$$

$$Q(A - BD^{-1}C) + (A - BD^{-1}C)^*Q + C^*(D^{-1})^*D^{-1}C = 0. \tag{7.23}$$

(b) Suppose the realization for G is weighted balanced, i.e.,

$$P = Q = \operatorname{diag}(\sigma_1 I_{s_1}, \ldots, \sigma_r I_{s_r}, \sigma_{r+1} I_{s_{r+1}}, \ldots, \sigma_N I_{s_N}) = \operatorname{diag}(\Sigma_1, \Sigma_2)$$

with $\sigma_1 > \sigma_2 > \ldots > \sigma_N \ge 0$ and let the realization of G be partitioned compatibly with Σ_1 and Σ_2 as

$$G(s) = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right].$$

Then

$$G_r(s) = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is stable and minimum phase. Furthermore

$$\|\Delta_{rel}\|_\infty \le \prod_{i=r+1}^{N} \left(1 + 2\sigma_i(\sqrt{1+\sigma_i^2} + \sigma_i)\right) - 1$$

$$\|\Delta_{mul}\|_\infty \le \prod_{i=r+1}^{N} \left(1 + 2\sigma_i(\sqrt{1+\sigma_i^2} + \sigma_i)\right) - 1.$$

Proof. Since the input weighting matrix $W_i = I$, it is obvious that the input weighted Gramian is given by P. Now the output weighted transfer matrix is given by

$$G^{-1}(G-D) = \left[\begin{array}{cc|c} A & 0 & B \\ -BD^{-1}C & A-BD^{-1}C & 0 \\ \hline D^{-1}C & D^{-1}C & 0 \end{array}\right] =: \left[\begin{array}{c|c} \bar{A} & \bar{B} \\ \hline \bar{C} & 0 \end{array}\right].$$

It is easy to verify that

$$\bar{Q} := \begin{bmatrix} Q & Q \\ Q & Q \end{bmatrix}$$

satisfies the following Lyapunov equation

$$\bar{Q}\bar{A} + \bar{A}^*\bar{Q} + \bar{C}^*\bar{C} = 0.$$

Hence Q is the output weighted Gramian.

The proof for part (b) is more involved and needs much more work. We refer readers to the references at the end of the chapter for details. □

In the above theorem, we have assumed that the system is square, we shall now extend the results to include non-square case. Let $G(s)$ be a minimum phase transfer matrix with a minimal realization

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

and assume that D has full row rank. Without loss of generality, we shall also assume that D is normalized such that $DD^* = I$. Let $D_\perp$ be a matrix with full row rank such that $\begin{bmatrix} D \\ D_\perp \end{bmatrix}$ is square and unitary.

Lemma 7.6 *A complex number $z \in \mathbb{C}$ is a zero of $G(s)$ if and only if z is an uncontrollable mode of $(A - BD^*C, BD_\perp^*)$.*

Proof. Since D has full row rank and $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ is a minimal realization, z is a transmission zero of $G(s)$ if and only if

$$\begin{bmatrix} A - zI & B \\ C & D \end{bmatrix}$$

does not have full row rank. Now note that

$$\begin{bmatrix} A - zI & B \\ C & D \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & [D^*, D_\perp^*] \end{bmatrix} \begin{bmatrix} I & 0 & 0 \\ -C & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

$$= \begin{bmatrix} A - BD^*C - zI & BD^* & BD_\perp^* \\ 0 & I & 0 \end{bmatrix}.$$

Then it is clear that

$$\begin{bmatrix} A - zI & B \\ C & D \end{bmatrix}$$

does not have full row rank if and only if

$$\begin{bmatrix} A - BD^*C - zI & BD_\perp^* \end{bmatrix}$$

does not have full row rank. By PBH test, this implies that z is a zero of $G(s)$ if and only if it is an uncontrollable mode of $(A - BD^*C, BD_\perp^*)$. □

Corollary 7.7 *There exists a matrix $\tilde{C}$ such that the augmented system*

$$G_a := \left[\begin{array}{c|c} A & B \\ \hline C_a & D_a \end{array}\right] = \left[\begin{array}{c|c} A & B \\ \hline C & D \\ \tilde{C} & D_\perp \end{array}\right] = \begin{bmatrix} G(s) \\ \tilde{G}(s) \end{bmatrix}$$

is minimum phase.

Proof. Note that the zeros of G_a are given by the eigenvalues of

$$A - B \begin{bmatrix} D \\ D_\perp \end{bmatrix}^{-1} \begin{bmatrix} C \\ \tilde{C} \end{bmatrix} = A - BD^*C - BD_\perp^*\tilde{C}.$$

Hence $\tilde{C}$ can always be chosen so that $A - BD^*C - BD_\perp^*\tilde{C}$ is stable. □

If the previous model reduction algorithms are applied to the augmented system G_a, the corresponding P and Q equations are given by

$$PA^* + AP + BB^* = 0$$

$$Q(A - BD_a^{-1}C_a) + (A - BD_a^{-1}C_a)^*Q + C_a^*(D_a^{-1})^*D_a^{-1}C_a = 0.$$

Moreover, we have

$$\begin{bmatrix} \hat{G}(s) \\ \hat{\tilde{G}}(s) \end{bmatrix} = \begin{bmatrix} G(s) \\ \tilde{G}(s) \end{bmatrix} (I + \Delta_{rel}), \quad \begin{bmatrix} G(s) \\ \tilde{G}(s) \end{bmatrix} = \begin{bmatrix} \hat{G}(s) \\ \hat{\tilde{G}}(s) \end{bmatrix} (I + \Delta_{mul})$$

and

$$\hat{G}(s) = G(s)(I + \Delta_{rel}), \quad G(s) = \hat{G}(s)(I + \Delta_{mul}).$$

However, there are in general infinitely many choices of $\tilde{C}$ and the model reduction results will in general depend on the specific choice. Hence an appropriate choice of $\tilde{C}$ is important. To motivate our choice, note that the equation for Q can be rewritten as

$$Q(A-BD^*C)+(A-BD^*C)^*Q-QBD_\perp^*D_\perp B^*Q+C^*C+(\tilde{C}-D_\perp B^*Q)^*(\tilde{C}-D_\perp B^*Q) = 0$$

A natural choice might be

$$\tilde{C} = D_\perp B^* Q.$$

The existence of a solution Q to the following so-called algebraic Riccati equation

$$Q(A - BD^*C) + (A - BD^*C)^*Q - QBD_\perp^* D_\perp B^* Q + C^*C = 0$$

such that

$$A - BD_a^{-1}C_a = A - BD^*C - BD_\perp^*\tilde{C} = A - BD^*C - BD_\perp^* D_\perp B^* Q$$

is stable will be shown in Chapter 13.

In the case where the model is not stable and /or is not minimum phase, the following procedure can be used: Let $G(s)$ be factorized as $G(s) = G_{ap}(s)G_{mp}(s)$ such that G_{ap} is an all-pass, i.e., $G_{ap}^\sim G_{ap} = I$, and G_{mp} is stable and minimum phase. Let $\hat{G}_{mp}$ be a relative/multiplicative reduced model of G_{mp} such that

$$\hat{G}_{mp} = G_{mp}(I + \Delta_{\mathrm{rel}})$$

and

$$G_{mp} = \hat{G}_{mp}(I + \Delta_{\mathrm{mul}}).$$

Then $\hat{G} := G_{ap}\hat{G}_{mp}$ has exactly the same right half plane poles and zeros as that of G and

$$\hat{G} = G(I + \Delta_{\mathrm{rel}})$$

$$G = \hat{G}(I + \Delta_{\mathrm{mul}}).$$

Unfortunately, this approach may be conservative if the transfer matrix has many non-minimum phase zeros or unstable poles.

An alternative relative/multiplicative model reduction approach, which does not require that the transfer matrix be minimum phase but does require solving an algebraic Riccati equation, is the so-called Balanced Stochastic Truncation (BST) method. Let $G(s) \in \mathcal{RH}_\infty$ be a square transfer matrix with a state space realization

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

and $\det(D) \neq 0$. Let $W(s) \in \mathcal{RH}_\infty$ be a minimum phase left spectral factor of $G(s)G^\sim(s)$, i.e,

$$W^\sim(s)W(s) = G(s)G^\sim(s).$$

Then $W(s)$ can be obtained as

$$W(s) = \left[\begin{array}{c|c} A & B_W \\ \hline C_W & D^* \end{array}\right]$$

with

$$\begin{aligned} B_W &= PC^* + BD^* \\ C_W &= D^{-1}(C - B_W^* X) \end{aligned}$$

where P is the controllability Gramian given by

$$AP + PA^* + BB^* = 0 \tag{7.24}$$

and X is the solution of a Riccati equation

$$XA_W + A_W^* X + XB_W(DD^*)^{-1}B_W^* X + C^*(DD^*)^{-1}C = 0 \tag{7.25}$$

with $A_W := A - B_W(DD^*)^{-1}C$ such that $A_W + B_W(DD^*)^{-1}B_W^* X$ is stable. The realization G is said to be a *balanced stochastic realization* if

$$P = X = \begin{bmatrix} \mu_1 I_{s_1} & & & \\ & \mu_2 I_{s_2} & & \\ & & \ddots & \\ & & & \mu_n I_{s_n} \end{bmatrix}$$

with $\mu_1 > \mu_2 > \ldots > \mu_n \geq 0$. μ_i is in fact the i-th Hankel singular value of the so-called "phase matrix" $(W^\sim(s))^{-1}G(s)$.

Theorem 7.8 *Let $G(s) \in \mathcal{RH}_\infty$ have the following balanced stochastic realization*

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right]$$

with $\det(D) \neq 0$ *and* $P = X = \operatorname{diag}(M_1, M_2)$ *where*

$$M_1 = \operatorname{diag}(\mu_1 I_{s_1}, \ldots, \mu_r I_{s_r}),\ M_2 = \operatorname{diag}(\mu_{r+1} I_{s_{r+1}}, \ldots, \mu_n I_{s_n}).$$

Then

$$\hat{G} = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is stable and

$$\left\| G^{-1}(G - \hat{G}) \right\|_\infty \leq \prod_{i=r+1}^{n} \frac{1+\mu_i}{1-\mu_i} - 1$$

$$\left\| \hat{G}^{-1}(G - \hat{G}) \right\|_\infty \leq \prod_{i=r+1}^{n} \frac{1+\mu_i}{1-\mu_i} - 1.$$

It can be shown that the balanced stochastic realization and the self-weighted balanced realization in Theorem 7.5 are the same and $\mu_i = \sigma_i/\sqrt{1+\sigma_i^2}$ if $G(s)$ is minimum phase. To illustrate, consider a third order stable and minimum phase transfer function

$$G(s) = \frac{s^3 + 2s^2 + 3s + 4}{s^3 + 3s^2 + 4s + 4}.$$

It is easy to show that the Hankel singular values of the phase function is given by

$$\mu_1 = 0.55705372196966, \quad \mu_2 = 0.53088390857035, \quad \mu_3 = 0.03715882438832,$$

and a first order BST approximation is given by

$$\hat{G} = \frac{s + 0.00375717470515}{s + 0.00106883052786}.$$

The relative approximation error is 2.51522024045904 and the error bound is

$$\prod_{i=2}^{3} \frac{1+\mu_i}{1-\mu_i} - 1 = 2.51522024046226.$$

7.4 Notes and References

The balanced model reduction method was first introduced by Moore [1981]. The stability properties of the reduced order model were shown by Pernebo and Silverman [1982]. The error bound for the balanced model reduction was shown by Enns [1984] and Glover [1984] subsequently gave an independent proof. The frequency weighted balanced model reduction method was also introduced by Enns [1984] from a somewhat different perspective. The error bounds for the relative and multiplicative approximations using the self-weighted balanced realization were shown by Zhou [1993]. The Balanced Stochastic Truncation (BST) method was proposed by Desai and Pal [1984] and generalized by Green [1988a,1988b] and many other people. The relative error bound for the Balanced Stochastic Truncation method was obtained by Green [1988a] and the multiplicative error bound for the BST was obtained by Wang and Safonov [1992]. It was also shown in Zhou [1993] that the frequency self-weighted method and the BST method are the same if the transfer matrix is minimum phase. Improved error bounds for the BST reduction were reported in Wang and Safonov [1990,1992] where it was claimed that the following error bounds hold

$$\left\| G^{-1}(G - \hat{G}) \right\|_\infty \le 2 \sum_{i=r+1}^{n} \frac{\mu_i}{1-\mu_i}$$

$$\left\| \hat{G}^{-1}(G - \hat{G}) \right\|_\infty \le 2 \sum_{i=r+1}^{n} \frac{\mu_i}{1-\mu_i}.$$

However, the example in the last section gives

$$2\sum_{i=2}^{3}\frac{\mu_i}{1-\mu_i} = 2.34052280324021$$

which is smaller than the actual error. Other weighted model reduction methods can be found in Al-Saggaf and Franklin [1988], Glover [1986,1989], Glover, Limebeer and Hung [1992], Hung and Glover [1986] and references therein. Discrete time balance model reduction results can be found in Al-Saggaf and Franklin [1987], Hinrichsen and Pritchard [1990], and references therein.